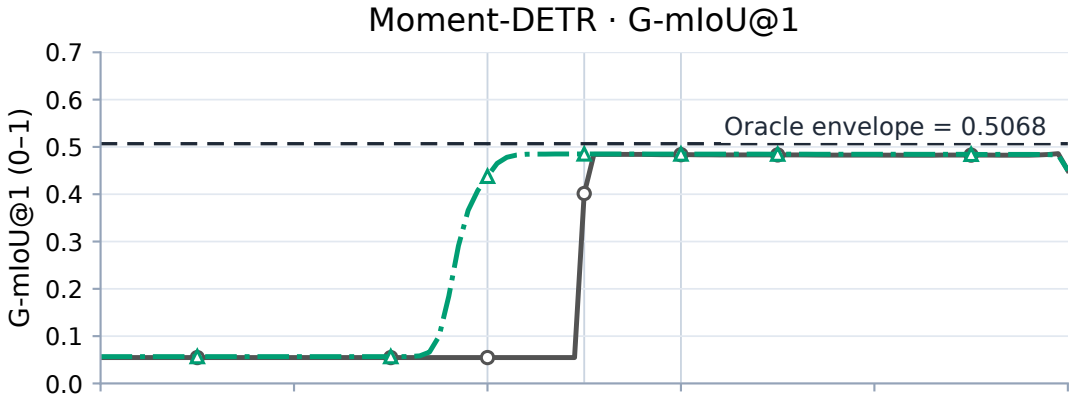
# G-mIoU sensitivity by backbone · mAP and changes versus GMR

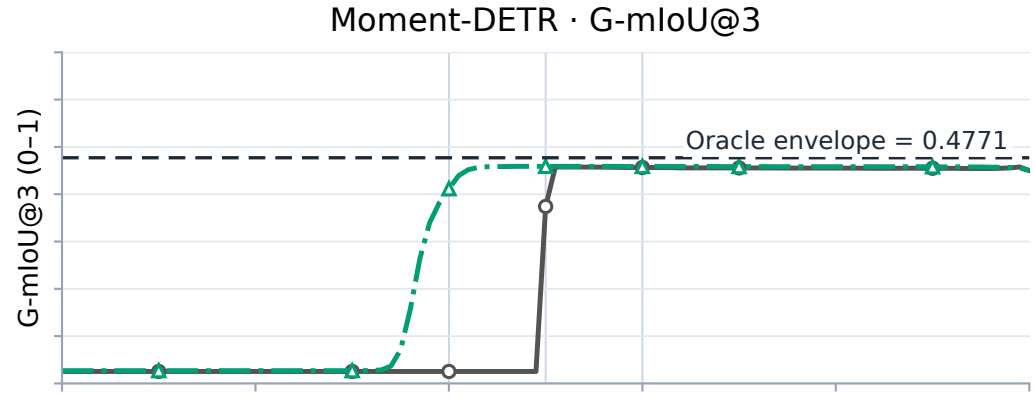
Standard-trained, fixed Val-selected checkpoints → SoccerGMR Full-Decodable Test (2970/2986 queries)

Historical GEQ protocol: 6 non-improving epochs; no Test selection or tuning

## Moment-DETR · G-mIoU@1



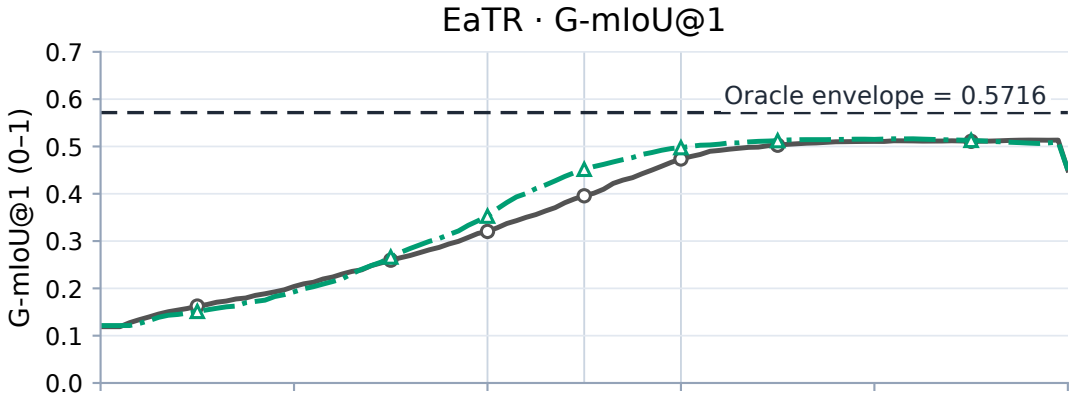

## Moment-DETR · G-mIoU@3



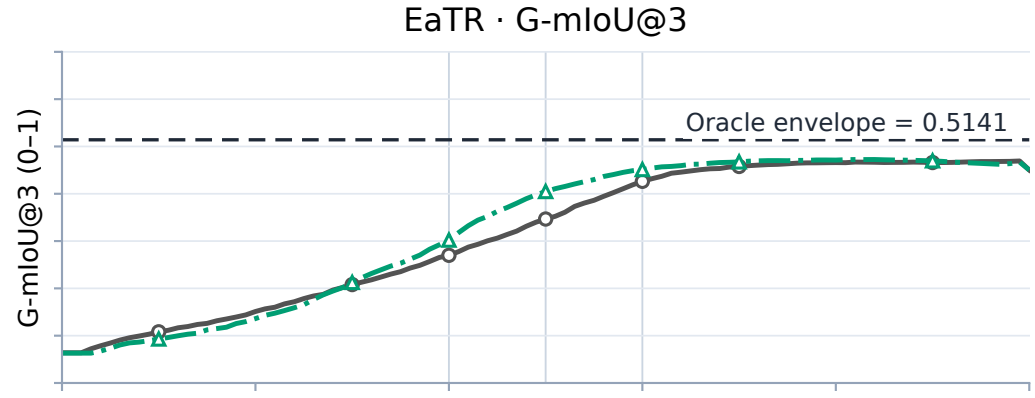

GMR baseline | mAP 5.47% — GMR + GEQ Full | mAP 5.47%

ΔmAP vs GMR: +0.00 pp — ΔG1 mean (.4/.5/.6): +15.60 pp — ΔG3 mean (.4/.5/.6): +15.77 pp

## EaTR · G-mIoU@1



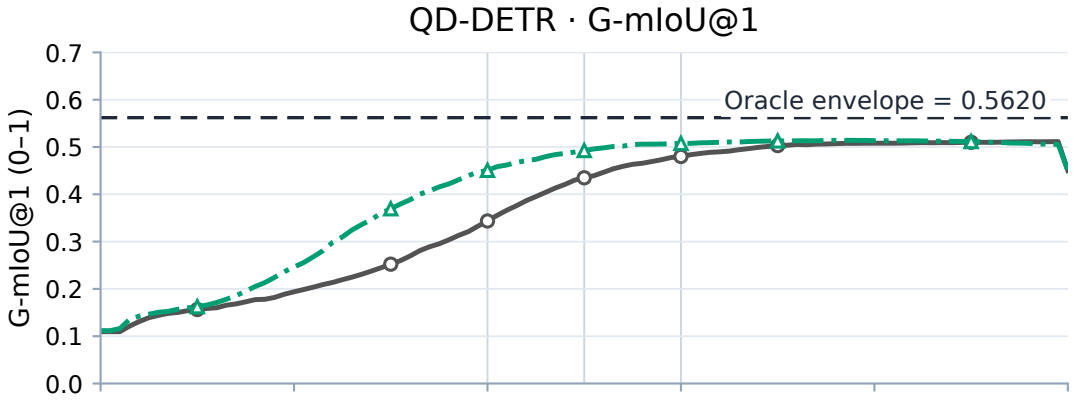

## EaTR · G-mIoU@3



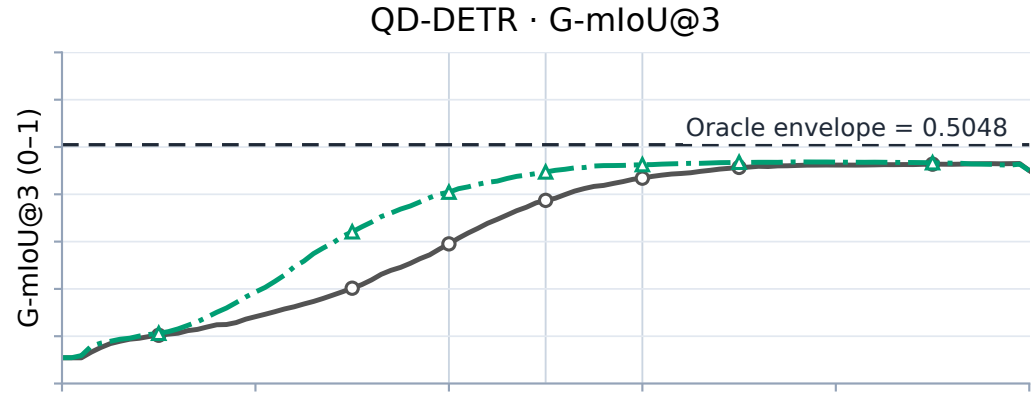

GMR baseline | mAP 15.19% — GMR + GEQ Full | mAP 15.26%

ΔmAP vs GMR: +0.07 pp — ΔG1 mean (.4/.5/.6): +3.73 pp — ΔG3 mean (.4/.5/.6): +3.84 pp

## QD-DETR · G-mIoU@1



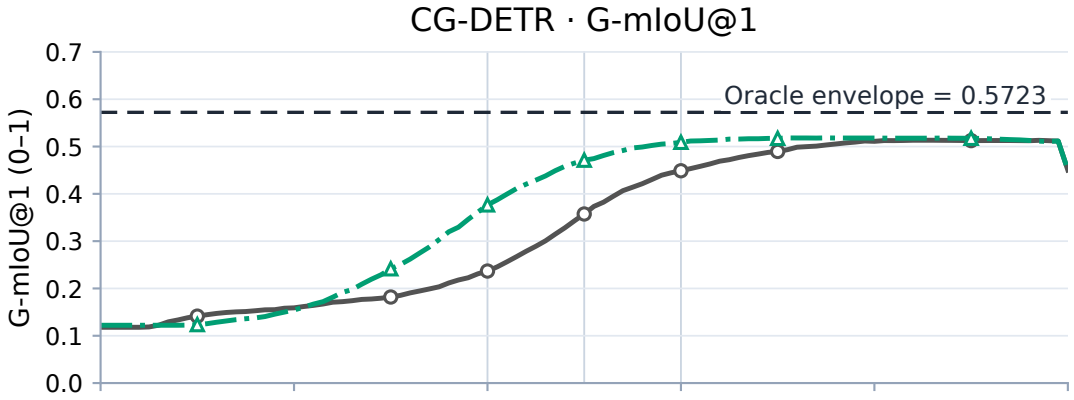

## QD-DETR · G-mIoU@3



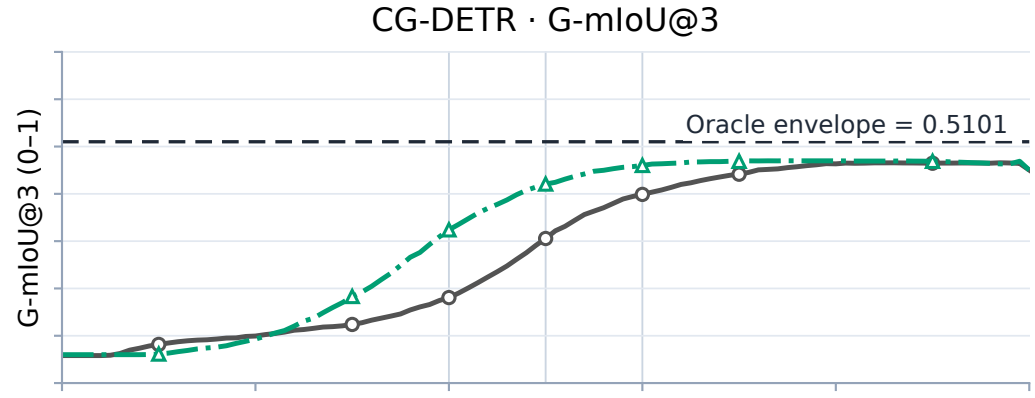

GMR baseline | mAP 12.65% — GMR + GEQ Full | mAP 13.22%

ΔmAP vs GMR: +0.57 pp — ΔG1 mean (.4/.5/.6): +6.40 pp — ΔG3 mean (.4/.5/.6): +6.59 pp

## CG-DETR · G-mIoU@1



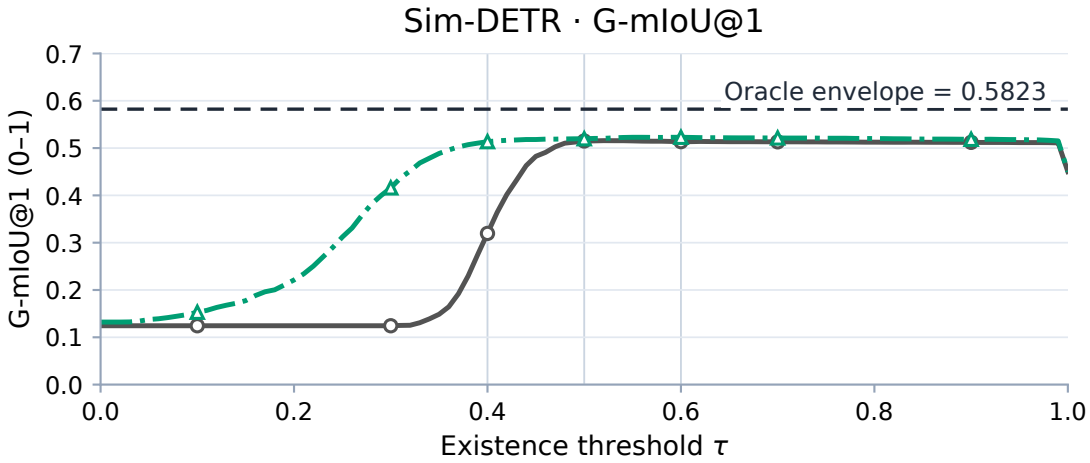

## CG-DETR · G-mIoU@3



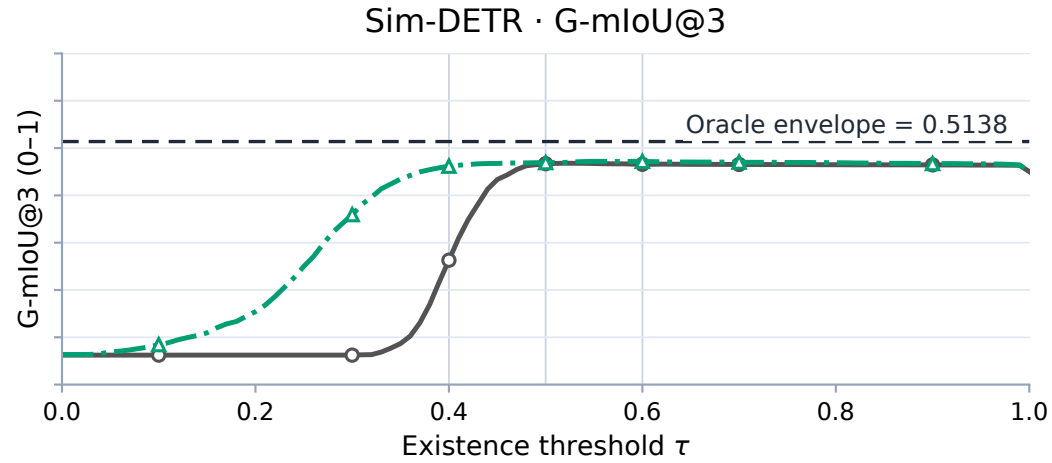

GMR baseline | mAP 14.67% — GMR + GEQ Full | mAP 14.85%

ΔmAP vs GMR: +0.18 pp — ΔG1 mean (.4/.5/.6): +10.45 pp — ΔG3 mean (.4/.5/.6): +10.65 pp

## Sim-DETR · G-mIoU@1




## Sim-DETR · G-mIoU@3




GMR baseline | mAP 19.18% — GMR + GEQ Full | mAP 20.58%

ΔmAP vs GMR: +1.40 pp — ΔG1 mean (.4/.5/.6): +6.93 pp — ΔG3 mean (.4/.5/.6): +6.90 pp

Oracle envelope: max of the two plotted methods with GT presence and each method's fixed candidates.

Curves: all 101 fixed thresholds. Δ: GEQ Full − matching GMR. G1/G3 means use τ = .4/.5/.6; pp = percentage points.

Legacy 75-token input limit retained: two long Full videos are truncated; 16 undecodable queries remain uncovered.